UNIT-4

16/10/14.

RECURRENCE RELATIONS.

Defination:

A recurrence relation for the sequence $a_0, a_1, a_2, \ldots a_{n-1}, a_n \ldots$ is an equation that relates a_n to some of its previous terms $a_0, a_1, a_2 \ldots a_{n-1}$.

A sequence is called solution of recurrence relation.

Defination:

A recurrence relation of any relation is an equation that expresse in terms of one or more previous terms of the sequence. namely $a_0, a_1, a_2, \ldots a_{n-1}$.

Eg:

$$a_n = n + a_{n-1}$$

$$a_n - 3a_{n-1} + 2a_{n-2} = 0.$$

Fibonacci recurrence relation:

Defn:

The recurrence relation $F_n = F_{n-1} + F_{n-2}$, $n \geq 2$ with initial conditions $F_0 = F_1 = 1$ is known as Fibonacci recurrence relation. (or) simply Fibonacci relation.

Eg:

Find Fibonacci (relation) series or Fibonacci sequence upto 5.

Sol.

Where $n = \{0, 1, 2, 3, 4, 5\}$.

Initial conditions are

$F_0 = 1$

$F_1 = 1$

$F_n = F_{n-1} + F_{n-2}$

$F_2 = F_{2-1} + F_{2-2}$

$= F_1 + F_1$

$= 1 + 1$

$= 2$

$F_3 = F_{3-1} + F_{3-2} = F_2 + F_1 = 2 + 1 = 3$

$F_4 = F_{4-1} + F_{4-2} = F_3 + F_2 = 3 + 2 = 5$

$F_5 = F_{5-1} + F_{5-2} = F_4 + F_3 = 5 + 3 = 8.$

∴ The Fibonacci series is

1 1 2 3 5 8

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Solving recurrence relation:

The process of finding an expression for the terms of sequence from its recurrence relation is called solving recurrence relation.

Defn:

The solution of recurrence relation is an expression for the general term a_n of the sequence $a_0, a_1, a_2, \ldots a_{n-1}$ satisfying the recurrence relation.

We have two methods for solving the recurrence relations.

- Solving recurrence relation by substitution.
- Solving recurrence relation by characteristic roots.

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Solving Recurrence relation by substitution:

In this method the given recurrence relation for a_n is used repeatedly to solve for a general expression for a_n in terms of n. This expression doesn't contains any other terms of the given sequence except those given by initial conditions.

Eg: 1 Solve the recurrence relation $a_n = a_{n-1} + f(n)$, $n \geq 1$ by substitution.

Sol: Given recurrence relation is

$$a_n = a_{n-1} + f(n)$$

clearly

$n=1$ $\quad a_1 = a_{1-1} + f(1) = a_0 + f(1)$

$n=2$ $\quad a_2 = a_{2-1} + f(2) = a_1 + f(2)$

$$= a_0 + f(1) + f(2)$$

$n=3$ $\quad a_3 = a_{3-1} + f(3) = a_2 + f(3)$

$$= a_0 + f(1) + f(2) + f(3)$$

$\vdots$

n $\quad a_n = a_0 + f(1) + f(2) + \ldots f(n)$.

Hence $a_n = a_0 + \sum_{k=1}^{n} f(k)$

2. Solve the recurrence relation $a_n = a_{n-1} + n^2$ where $a_0 = 7$ by substitution.

Sol: Given recurrence relation is

$$a_n = a_{n-1} + n^2$$

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Clearly

$n=1$ $\quad a_1 = a_{1-1} + 1^2 = a_0 + 1^2$

$n=2$ $\quad a_2 = a_{2-1} + 2^2 = a_1 + 2^2 = a_0 + 1^2 + 2^2$

$n=3$ $\quad a_3 = a_{3-1} + 3^2 = a_2 + 3^2 = a_0 + 1^2 + 2^2 + 3^2$

$\vdots$

n $\quad a_n = a_0 + 1^2 + 2^2 + 3^2 + \cdots + n^2$

Hence $\quad a_n = a_0 + \sum_{k=1}^{n} k^2$

since $a_0 = 7$

$$a_n = 7 + \sum_{k=1}^{n} k^2$$

3. Solve the recurrence relation $a_n = a_{n-1} + n^3$, $a_0 = 5$ by substitution.

Sol. Given recurrence relation is

$$a_n = a_{n-1} + n^3.$$

Clearly

$n=1$ $\quad a_1 = a_{1-1} + 1^3 = a_0 + 1^3$

$n=2$ $\quad a_2 = a_{2-1} + 2^3 = a_1 + 2^3 = a_0 + 1^3 + 2^3$

$n=3$ $\quad a_3 = a_{3-1} + 3^3 = a_2 + 3^3 = a_0 + 1^3 + 2^3 + 3^3$

$\vdots$

n $\quad a_n = a_0 + 1^3 + 2^3 + 3^3 + \cdots + n^3$

Hence $\quad a_n = a_0 + \sum_{k=1}^{n} k^3$

since $a_0 = 5$

$$a_n = 5 + \sum_{k=1}^{n} k^3.$$

4. Solve the recurrence relation $a_n = a_{n-1} + 3^n$, $a_0 = 1$ by substitution.

Sol. Given recurrence relation is

$$a_n = a_{n-1} + 3^n.$$

Clearly

$n=1$ $\quad a_1 = a_{1-1} + 3^1 = a_0 + 3^1$

$n=2$ $\quad a_2 = a_{2-1} + 3^2 = a_1 + 3^2 = a_0 + 3^1 + 3^2$

$n=3$ $\quad a_3 = a_{3-1} + 3^3 = a_2 + 3^3 = a_0 + 3^1 + 3^2 + 3^3$

$\vdots$

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n $\quad a_n = a_0 + 3^1 + 3^2 + 3^3 + \cdots + 3^n$.

Hence

$$a_n = a_0 + \sum_{k=1}^{n} 3^k$$

Since $a_0 = 1$

$$a_n = 1 + \sum_{k=1}^{n} 3^k.$$

5. Solve the recurrence relation $a_n = a_{n-1} + n$, $a_0 = 2$ by substitution.

Sol. Given recurrence relation is

$$a_n = a_{n-1} + n.$$

Clearly

$n=1$ $\quad a_1 = a_{1-1} + 1 = a_0 + 1$

$n=2$ $\quad a_2 = a_{2-1} + 2 = a_1 + 2 = a_0 + 1 + 2$

$n=3$ $\quad a_3 = a_{3-1} + 3 = a_2 + 3 = a_0 + 1 + 2 + 3$

⋮

n $\quad a_n = a_0 + 1 + 2 + 3 + \cdots + n$

Hence

$$a_n = a_0 + \sum_{k=1}^{n} k$$

Since $a_0 = 2$

$$a_n = 2 + \sum_{k=1}^{n} k.$$

Given recurrence relation is

$$a_n = a_{n-1} + n$$

Clearly

$n=1$ $\quad a_1 = a_{1-1} + 1 = a_0 + 1$

$n=2$ $\quad a_2 = a_{2-1} + 2 = a_1 + 2 = a_0 + 1 + 2$
$= a_0 + 3$

$n=3$ $\quad a_3 = a_{3-1} + 3 = a_2 + 3 = a_0 + 3 + 3$
$= a_0 + 6.$

$n=4$ $\quad a_4 = a_{4-1} + 4 = a_0 + 6 + 4$
$= a_0 + 10$

⋮

n $\quad a_n = a_0 + 1 + 3 + 6 + 10$

$$a_n = a_0 + \frac{n(n+1)}{2}.$$

Since $a_0 = 2$.

$$a_n = 2 + \frac{n(n+1)}{2}.$$

6. Solve the recurrence relation $a_n = a_{n-1} + 2n + 1$, $a_0 = 1$ by substitution.

Sol. Given recurrence relation is

$$a_n = a_{n-1} + 2n + 1$$

Clearly

$n=1$ $\quad a_1 = a_{1-1} + 2(1) + 1 = a_0 + 2 + 1 = a_0 + 3.$

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$n=2$ $a_2 = a_{2-1} + 2(2) + 1 = a_1 + 4 + 1 = a_0 + 5$

$= a_0 + 3 + 5 = a_0 + 8$

$n=3$ $a_3 = a_{3-1} + 2(3) + 1 = a_2 + 6 + 1 = a_2 + 7$

$= a_0 + 8 + 7 = a_0 + 15$

$n=4$ $a_4 = a_{4-1} + 2(4) + 1 = a_3 + 8 + 1$

$= a_0 + 15 + 8 + 1$

$= a_0 + 24.$

3, 8, 15, 24, - - -

$n(n+2)$

$n=1 \quad 1(1+2) = 3$

$n=2 \quad 2(2+2) = 8$

$n=3 \quad 3(3+2) = 15$

$n=4 \quad 4(4+2) = 24$

$n \quad a_n = a_0 + n(n+2)$

Since $a_0 = 1$

$a_n = 1 + n(n+2)$

7. Solve the recurrence relation $a_n = a_{n-1} + 3n^2 + 3n + 1$, $a_0 = 1$ by substitution.

Sol. Given recurrence relation is

$a_n = a_{n-1} + 3n^2 + 3n + 1$

$n=1$ $a_1 = a_{1-1} + 3(1)^2 + 3(1) + 1 = a_0 + 3 + 3 + 1$

$= a_0 + 7 = 1 + 7 = 8$

$n=2$ $a_2 = a_{2-1} + 3(2)^2 + 3(2) + 1 = a_1 + 12 + 6 + 1$

$= a_1 + 19 = a_0 + 7 + 19 = a_0 + 26$

$= 1 + 26 = 27$

$n=3$ $a_3 = a_{3-1} + 3(3)^2 + 3(3) + 1 = a_2 + 27 + 9 + 1$

$= a_0 + 26 + 27 + 9 + 1$

$= a_0 + 63 = 1 + 63 = 64$

8, 27, 64, - - -

$(n+1)^3$

$n=1 \quad (1+1)^3 = 2^3 = 8$

$n=2 \quad (2+1)^3 = 3^3 = 27$

$n=3 \quad (3+1)^3 = 4^3 = 64$

$n \quad a_n = (n+1)^3$

8. Solve the recurrence relation $a_n = a_{n-1} + \frac{1}{n(n+1)}$, $a_0 = 1$ by substitution.

Sol. Given recurrence relation is

$a_n = a_{n-1} + \frac{1}{n(n+1)}$

$n=1$ $a_1 = a_{1-1} + \frac{1}{1(1+1)} = a_0 + \frac{1}{2}$

$n=2$ $a_2 = a_{2-1} + \frac{1}{2(2+1)} = a_1 + \frac{1}{6} = a_0 + \frac{1}{2} + \frac{1}{6}$

$= a_0 + \frac{4}{6} = a_0 + \frac{2}{3}$

$n=3$ $a_3 = a_{3-1} + \frac{1}{3(3+1)} = a_2 + \frac{1}{12} = a_0 + \frac{1}{2} + \frac{1}{6} + \frac{1}{12}$

$= a_0 + \frac{9}{12} = a_0 + \frac{3}{4}$

$n=4$ $a_4 = a_{4-1} + \frac{1}{4(4+1)} = a_3 + \frac{1}{20} = a_0 + \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20}$

$= a_0 + \frac{48}{60} = a_0 + \frac{4}{5}$

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8. Solve the recurrence relation

$a_n = a_{n-1} + \frac{1}{n(n+1)}$, $a_0 = 1$ by Substitution.

Sol.

Given recurrence relation is

$$a_n = a_{n-1} + \frac{1}{n(n+1)}$$

$n=1 \Rightarrow a_{1-1} + \frac{1}{1(1+1)}$

$= a_0 + 1/2 = 1 + 1/2 = 3/2$

$n=2 \Rightarrow a_{2-1} + \frac{1}{2(2+1)}$

$= a_1 + \frac{1}{2(3)} = a_1 + 1/6 = a_0 + 1/2 + 1/6$

$= a_0 + 2/3 = 1 + 2/3 = 5/3$

$n=3 \Rightarrow a_{3-1} + \frac{1}{3(3+1)}$

$\Rightarrow a_2 + 1/12$

$\Rightarrow a_0 + 2/3 + 1/12$

$= a_0 + 9/12$

$= a_0 + 3/4$

$= 1 + 3/4$

$= 7/4$

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Solving the recurrence relation by Characteristic roots:

In this method to solve recurrence relation of degree k for this we require the defination of characteristic equation of recurrence relation.

Defination:

let $a_n + c_1 a_{n-1} + \cdots + c_k a_{n-k} = 0$,

$n \geq k$, $c_k \neq 0$ be a recurrence relation of degree k then the eqn is

$r^k + c_1 r^{k-1} + \cdots + c_k = 0$ is said to be characteristic equation.

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Eg:

1. Solve $a_n - 7a_{n-1} + 12a_{n-2} = 0$ by using characteristic roots.

Sol. Given recurrence relation is

$$a_n - 7a_{n-1} + 12a_{n-2} = 0.$$

Then the characteristic equation is

$$r^2 - 7r + 12 = 0.$$

$$r^2 - 3r - 4r + 12 = 0$$

$$\Rightarrow r(r-3) - 4(r-3) = 0$$

$$\Rightarrow (r-3)(r-4) = 0$$

$$\Rightarrow r = 3, 4.$$

$\therefore$ The general solution is

$$a_n = c_1 3^n + c_2 4^n.$$

2. Solve $a_n - 5a_{n-1} + 6a_{n-2} = 0$ by using characteristic roots, where $a_0 = 2$ and $a_1 = 5$.

Sol. Given recurrence relation is

$$a_n - 5a_{n-1} + 6a_{n-2} = 0.$$

The characteristic equation is

$$r^2 - 5r + 6 = 0$$

$$r^2 - 3r - 2r + 6 = 0.$$

$$r(r-3) - 2(r-3) = 0$$

$$(r-3)(r-2) = 0$$

$$r = 2, 3$$

$\therefore$ The general solution is

$$a_n = c_1 2^n + c_2 3^n.$$

By initial conditions

$$\therefore a_0 = c_1 2^0 + c_2 3^0 = 2$$

$$\Rightarrow c_1 + c_2 = 2 \longrightarrow (1)$$

$$a_1 = c_1 2^1 + c_2 3^1 = 5$$

$$\Rightarrow 2c_1 + 3c_2 = 5 \longrightarrow (2)$$

Solving (1) & (2)

$$\begin{array}{r} (c_1 + c_2 = 2) \times 2 \\ (2c_1 + 3c_2 = 5) \times 1 \\ \hline 2c_1 + 2c_2 = 4 \\ 2c_1 + 3c_2 = 5 \\ \hline -c_2 = -1 \\ \hline +c_2 = +1 \end{array}$$

$$c_2 = 1$$

$$c_1 + 1 = 2$$

$$c_1 = 1$$

$\therefore c_1 = 1, c_2 = 1$

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then the solution is

$$a_n = 1 \cdot 2^n + 1 \cdot 3^n$$

$$a_n = 2^n + 3^n.$$

3. Solve $a_n - 7a_{n-1} + 10a_{n-2} = 0$ by using characteristic roots, where $a_0 = 10$ and $a_1 = 41$.

Sol: Given recurrence relation is

$$a_n - 7a_{n-1} + 10a_{n-2} = 0.$$

The characteristic equation is

$$\gamma^2 - 7\gamma + 10 = 0$$

$$\gamma^2 - 2\gamma - 5\gamma + 10 = 0$$

$$\gamma(\gamma-2) - 5(\gamma-2) = 0$$

$$(\gamma-2)(\gamma-5) = 0$$

$$\gamma = 2, 5$$

∴ The general solution is

$$a_n = c_1 2^n + c_2 5^n$$

By initial conditions

$$a_0 = c_1 2^0 + c_2 5^0 = 10$$

$$c_1 + c_2 = 10 \longrightarrow (1)$$

$$a_1 = c_1 2^1 + c_2 5^1 = 41$$

$$2c_1 + 5c_2 = 41 \longrightarrow (2)$$

$(c_1 + c_2 = 10) 2$

$(2c_1 + 5c_2 = 41) 1$

$2c_1 + 2c_2 = 20$

$2c_1 + 5c_2 = 41$

$-3c_2 = -21$

$c_2 = 21/3$

$c_2 = 7$

$c_1 + c_2 = 10$

$c_1 + 7 = 10$

$c_1 = 3$

∴ $c_1 = 3, c_2 = 7$

then the solution is

$$a_n = 3 \cdot 2^n + 7 \cdot 5^n.$$

4. Solve $a_n - 6a_{n-1} + 9a_{n-2} = 0$ by using characteristic roots, where $a_0 = 2$ and $a_1 = 9$.

Sol: The given recurrence relation is

$$a_n - 6a_{n-1} + 9a_{n-2} = 0.$$

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The characteristic equation is

$$r^2 - 6r + 9 = 0$$

$$r^2 - 3r - 3r + 9 = 0$$

$$r(r-3) - 3(r-3) = 0$$

$$(r-3)(r-3) = 0$$

$$r = 3, 3.$$

∴ The general solution is

$$a_n = c_1 3^n + c_2 \cdot n \cdot 3^n.$$

By initial conditions

$$\therefore a_0 = c_1 3^0 + c_2 \cdot 0 \cdot 3^0$$

$$= c_1 = 2$$

$$a_1 = c_1 3^1 + c_2 \cdot 1 \cdot 3^1$$

$$\Rightarrow 3c_1 + 3c_2 = 9$$

$$3(2) + 3c_2 = 9$$

$$3c_2 = 9 - 6$$

$$c_2 = 3/3$$

$$c_2 = 1$$

$$\therefore c_1 = 2, c_2 = 1$$

then the solution is

$$a_n = 2 \cdot 3^n + 1 \cdot n \cdot 3^n.$$

5. Find characteristic roots of given equation. $r^3 - 7r^2 + 16r - 12 = 0$.

Sol.

$$r^3 - 7r^2 + 16r - 12 = 0$$

$r = 1 \Rightarrow 1 - 7 + 16 - 12 = 0$

$-2 \neq 0$

$r = -1 \Rightarrow -1 - 7 + 16 - 12 = 0$

$-36 \neq 0$

$r = 2 \Rightarrow 8 - 28 + 32 - 12 = 0$

$0 = 0.$

Synthetic division method.

$r=2$	1	−7	16	−12
	0	2	−10	12
	1	−5	6	0

$r = 2, r^2 - 5r + 6 = 0$

$r = 2, r^2 - 2r - 3r + 6 = 0$

$r = 2, r(r-2) - 3(r-2) = 0$

$r = 2, (r-2)(r-3) = 0$

$r = 2, 2, 3.$

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21/10/14.

1. Solve the recurrence relation $a_n - 7a_{n-1} + 16a_{n-2} - 12a_{n-3} = 0$, where $a_0 = 1$, $a_1 = 4$ and $a_2 = 8$.

Sol. The characteristic equation of given relation

$$r^3 - 7r^2 + 16r - 12 = 0$$

factors of constant value 12 are

$r = 1$

$1 - 7 + 16 - 12 = 0$ $\quad$ 1, 2, 3, 4, 6, 12

$-2 \neq 0$

$r = 2$

$8 - 28 + 32 - 12 = 0$

$0 = 0$.

$\therefore 2$ is one root.

Synthetic division

$r=2$	1	-7	16	-12
	0	2	-10	12
	1	-5	6	0

$r = 2, \quad r^2 - 5r + 6 = 0$

$r = 2, \quad r^2 - 2r - 3r + 6 = 0$

$r = 2, \quad r(r-2) - 3(r-2) = 0$

$r = 2, \quad (r-2)(r-3) = 0$

$r = 2, 2, 3.$

General solution is $a_n = c_1 2^n + c_2 n 2^n + c_3 3^n$.

By initial conditions

$$a_0 = c_1 2^0 + c_2 \cdot 0 \cdot 2^0 + c_3 3^0$$

$$= c_1 + c_3 = 1 \rightarrow (1)$$

$$a_1 = c_1 2^1 + c_2 \cdot 1 \cdot 2^1 + c_3 3^1$$

$$= 2c_1 + 2c_2 + 3c_3 = 4 \rightarrow (2)$$

$$a_2 = c_1 2^2 + c_2 \cdot 2 \cdot 2^2 + c_3 \cdot 3^2$$

$$= 4c_1 + 8c_2 + 9c_3 = 8 \rightarrow (3)$$

Solving (1), (2) & (3) we get

from (1) & (2)

$(c_1 + 0 \cdot c_2 + c_3 = 1) 2$

$(2c_1 + 2c_2 + 3c_3 = 4) 1$

$2c_1 + 0c_2 + 2c_3 = 2$

$2c_1 + 2c_2 + 3c_3 = 4$

$-2c_2 - c_3 = -2$

~~$-(2c_3 + c_3) = -2$~~

$-2c_2 - c_3 = -2$

$-(2c_2 + c_3) = -2$

$2c_2 + c_3 = 2 \rightarrow (4)$

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from (2) & (3)

$2c_1 + 2c_2 + 3c_3 = 4$)2

$4c_1 + 8c_2 + 9c_3 = 8$)1

$4c_1 + 4c_2 + 6c_3 = 8$

$4c_1 + 8c_2 + 9c_3 = 8$

$-4c_2 - 3c_3 = 0$

$-4c_2 - 3c_3 = 0$

$-(4c_2 + 3c_3) = 0$

$4c_2 + 3c_3 = 0 \rightarrow (5)$

from (4) & (5)

$2c_2 + c_3 = 2$)2

$4c_2 + 3c_3 = 0$)1

$4c_2 + 2c_3 = 4$

$4c_2 + 3c_3 = 0$

$-c_3 = 4$

$c_3 = -4$

Sub

$c_1 + c_3 = 1$

$c_1 - 4 = 1$

$c_1 = 5$

$2c_1 + 2c_2 + 3c_3 = 4$

$2(5) + 2c_2 + 3(-4) = 4$

$10 + 2c_2 - 12 = 4$

$2c_2 = 4 + 12 - 10$

$2c_2 = 6$

$c_2 = 3$.

Final solution

$a_n = 5 \cdot 2^n + 3 \cdot n \cdot 2^n + (-4) \cdot 3^n$.

2. Solve the recurrence relation

$a_n - 8a_{n-1} + 21a_{n-2} - 18a_{n-3} = 0$, where

$a_0 = 8$, $a_1 = 18$, $a_2 = 24$.

Sol. The characteristic equation of given relation

$r^3 - 8r^2 + 21r - 18 = 0$.

factors of constant value 18 are

1, 2, 3, 6, 9, 18.

$r = 1$

$1 - 8 + 21 - 18 = 0$

$-4 \neq 0$

$r = 2$ $8 - 32 + 42 - 18 = 0$

$0 = 0$.

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$\therefore 2$ is one root

Synthetic division.

$r=2$	1	−8	21	−18
	0	2	−12	18
	1	−6	9	0

$r=2,\ r^2-6r+9=0$

$r=2,\ r^2-3r-3r+9=0$

$r=2,\ r(r-3)-3(r-3)=0$

$r=2,\ (r-3)(r-3)=0$

$r=2, 3, 3.$

General solution is

$$a_n = c_1 \cdot 2^n + c_2 \cdot 3^n + c_3 \cdot n \cdot 3^n$$

By initial conditions.

$$a_0 = c_1 \cdot 2^0 + c_2 \cdot 3^0 + c_3 \cdot 0 \cdot 3^0$$

$$= c_1 + c_2 = 8 \longrightarrow (1.)$$

$$a_1 = c_1 \cdot 2^1 + c_2 \cdot 3^1 + c_3 \cdot 1 \cdot 3^1$$

$$= 2c_1 + 3c_2 + 3c_3 = 18 \longrightarrow (2.)$$

$$a_2 = c_1 \cdot 2^2 + c_2 \cdot 3^2 + c_3 \cdot 2 \cdot 3^2$$

$$= 4c_1 + 9c_2 + 18c_3 = 24 \longrightarrow (3.)$$

Solving (1.) & (2.)

$$(c_1 + c_2 + 0 \cdot c_3 = 8) \times 2$$

$$(2c_1 + 3c_2 + 3c_3 = 18) \times 1$$

$$2c_1 + 2c_2 + 0 \cdot c_3 = 16$$

$$2c_1 + 3c_2 + 3c_3 = 18$$

$$-c_2 - 3c_3 = -2$$

$$-c_2 - 3c_3 = -2$$

$$-(c_2 + 3c_3) = -2$$

$$c_2 + 3c_3 = 2 \longrightarrow (4.)$$

Solving (2.) & (3.)

$$(2c_1 + 3c_2 + 3c_3 = 18) \times 2$$

$$(4c_1 + 9c_2 + 18c_3 = 24) \times 1$$

$$4c_1 + 6c_2 + 6c_3 = 36$$

$$4c_1 + 9c_2 + 18c_3 = 24$$

$$-3c_2 - 12c_3 = 12$$

$$-3c_2 - 12c_3 = 12$$

$$-(3c_2 + 12c_3) = 12$$

$$3c_2 + 12c_3 = -12 \longrightarrow (5.)$$

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Solving (4) & (5)

$(C_2 + 3C_3 = 2)3$

$(3C_2 + 12C_3 = -12)1$

$3C_2 + 9C_3 = 6$

$3C_2 + 12C_3 = -12$

$-3C_3 = 18$

$C_3 = -6.$

$C_2 + 3C_3 = 2$

$C_2 + 3(-6) = 2$

$C_2 - 18 = 2$

$C_2 = 20$

$C_1 + C_2 = 8$

$C_1 + 20 = 8$

$C_1 = -12$

$C_1 = -12,\ C_2 = 20,\ C_3 = -6.$

Final solution

$a_n = -12 \cdot 2^n + 20 \cdot 3^n + (-6) \cdot n \cdot 3^n.$

Solving (4) & (5)

$(C_2 + 3C_3 = 2)3$

$(3C_2 + 12C_3 = -12)1$

$3C_2 + 9C_3 = 6$

$3C_2 + 12C_3 = -12$

$-3C_3 = 18$

$C_3 = -6.$

$C_2 + 3C_3 = 2$

$C_2 + 3(-6) = 2$

$C_2 - 18 = 2$

$C_2 = 20$

$C_1 + C_2 = 8$

$C_1 + 20 = 8$

$C_1 = -12$

$C_1 = -12,\ C_2 = 20,\ C_3 = -6.$

Final solution

$a_n = -12 \cdot 2^n + 20 \cdot 3^n + (-6) \cdot n \cdot 3^n.$

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3. Solve the recurrence relation

$a_n - 6a_{n-1} + 12a_{n-2} - 8a_{n-3} = 0$, where

$a_0 = 8$

$a_1 = 16$

$a_2 = 15$

Sol. The characteristic eqn of given relation is

$r^3 - 6r^2 + 12r - 8 = 0$.

factors of constant value are

1, 2, 4, 8.

$r = 1$

$1 - 6 + 12 - 8 = 0$

$-1 \neq 0$

$r = 2$

$8 - 24 + 24 - 8 = 0$

$0 = 0$.

$\therefore$ 2 is one root

Synthetic division

$r=2$	1	−6	12	−8
	0	2	−8	8
	1	−4	4	0

$r = 2, r^2 - 4r + 4 = 0$

$r = 2, r^2 - 2r - 2r + 4 = 0$

$r = 2, r(r-2) - 2(r-2) = 0$

$r = 2, (r-2)(r-2) = 0$

$r = 2, 2, 2$

General solution is

$a_n = c_1 \cdot 2^n + c_2 \cdot n \cdot 2^n + c_3 \cdot n^2 \cdot 2^n$.

By initial conditions

$a_0 = c_1 \cdot 2^0 + c_2 \cdot 0 \cdot 2^0 + c_3 \cdot 0^2 \cdot 2^0$

$= c_1 + 0 + 0$

$= c_1 = 8 \longrightarrow (1)$

$a_1 = c_1 \cdot 2^1 + c_2 \cdot 1 \cdot 2^1 + c_3 \cdot 1^2 \cdot 2^1$

$= 2c_1 + 2c_2 + 2c_3 = 16 \longrightarrow (2)$

a_2

$a_2 = c_1 \cdot 2^2 + c_2 \cdot 2 \cdot 2^2 + c_3 \cdot 2^2 \cdot 2^2$

$= 4c_1 + 8c_2 + 16c_3 = 15 \longrightarrow (3)$

$2c_1 + 2c_2 + 2c_3 = 16$

$2(8) + 2c_2 + 2c_3 = 16$

$16 + 2c_2 + 2c_3 = 16$

$2c_2 + 2c_3 = 0$

$2(c_2 + c_3) = 0$

$c_2 + c_3 = 0 \longrightarrow (4)$

$4c_1 + 8c_2 + 16c_3 = 15$

$4(8) + 8c_2 + 16c_3 = 15$

$32 + 8c_2 + 16c_3 = 15$

$8c_2 + 16c_3 = -17$

$\longrightarrow (5)$

Solving (4) & (5)

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$(C_2 + C_3 = 0)\ 8$

$(8C_2 + 16C_3 = -17)\ 1$

$8C_2 + 8C_3 = 0$

$8C_2 + 16C_3 = -17$

$-8C_3 = 17$

$C_3 = -17/8.$

$C_2 + C_3 = 0$

$C_2 - 17/8 = 0$

$C_2 = 17/8$

$C_1 = 8,\ C_2 = 17/8,\ C_3 = -17/8.$

Final solution is

$a_n = 8 \cdot 2^n + 17/8 \cdot n \cdot 2^n + (-17/8) \cdot n^2 \cdot 2^n.$

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Example:

1. Solve $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$.

Sol. The characteristic equation of given relation is

$r^3 - r^2 - 2r = 0$

$r(r^2 - r - 2) = 0$

$r(r^2 - 2r + r - 2) = 0$

$r(r(r-2) + 1(r-2)) = 0$

$r((r-2)(r+1)) = 0$

$r = 0, 2, -1$

$r = 0, -1, 2.$

General solution is

~~$a_n = c_1 \cdot 2^n + c_2 \cdot 2^n + c_3 \cdot 2^n$~~

~~By initial con~~

~~$a_n = c_1 \cdot 2^n + c_2 \cdot (-1)^n + c_3 \cdot 2^n$~~

$a_n = c_1(-1)^n + c_2 \cdot 2^n$

By initial conditions.

$a_0 = c_1(-1)^0 + c_2 \cdot 2^0 = 2$

$\Rightarrow c_1 + c_2 = 2 \rightarrow (1)$

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$a_1 = c_1(-1)^1 + c_2 \cdot 2^1 = 7$

$= -c_1 + 2c_2 = 7 \rightarrow (2)$

From (1) & (2)

$$c_1 + c_2 = 2$$
$$-c_1 + 2c_2 = 7$$
$$3c_2 = 9$$
$$c_2 = 3$$

$c_1 + c_2 = 2$

$c_1 + 3 = 2$

$c_1 = -1$

Final solution is

$a_n = (-1)(-1)^n + 3 \cdot 2^n$.

2. Solve $a_n - 3a_{n-1} - 4a_{n-2} = 3^n$ with $a_0 = 1$ and $a_1 = 9$

Sol. Given that

$a_n - 3a_{n-1} - 4a_{n-2} = 3^n$

L.H.S

$a_n - 3a_{n-1} - 4a_{n-2} = 0$.

Characteristic equation is

$r^2 - 3r - 4 = 0$

$r^2 - 4r + r - 4 = 0$

$r(r-4) + 1(r-4) = 0$

$(r+1)(r-4) = 0$

$r = -1, 4$.

The homogeneous solution is

$a_n(h) = c_1(-1)^n + c_2(4)^n$.

R.H.S

3^n

To obtain particular solution we take as $d \cdot 3^n$

Substitute $d \cdot 3^n$ for a_n

$d \cdot 3^n - 3 \cdot d \cdot 3^{n-1} - 4 \cdot d \cdot 3^{n-2} = 3^n$

$n = 2$

$d \cdot 3^2 - 3 \cdot d \cdot 3^{2-1} - 4 \cdot d \cdot 3^{2-2} = 3^2$

$9d - 9d - 4d = 9$

$d = (-9/4)$

The particular solution is

$a_n(P) = \left(\frac{-9}{4}\right) 3^n$

$a_n = c_1(-1)^n + c_2(4)^n - \frac{9}{4} \cdot 3^n$

By initial conditions

$a_0 = c_1(-1)^0 + c_2(4)^0 - \frac{9}{4} 3^0 = 1$

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$c_1 + c_2 - 9/4 = 1$

$c_1 + c_2 = 1 + 9/4$

$c_1 + c_2 = 13/4 \rightarrow (1)$

$a_1 = c_1(-1)^1 + c_2(4)^1 - \frac{9}{4} \cdot 3^1$

$= -c_1 + 4c_2 - \frac{27}{4} = 2$

$-c_1 + 4c_2 = 2 + \frac{27}{4}$

$-c_1 + 4c_2 = \frac{35}{4} \rightarrow (2)$

From (1) & (2)

$c_1 + c_2 = 13/4$

$-c_1 + 4c_2 = 35/4$

$5c_2 = 48/4$

$5c_2 = 12$

$c_2 = 12/5$

$c_1 + c_2 = 13/4$

$c_1 + 12/5 = 13/4$

$c_1 = \frac{13}{4} - \frac{12}{5}$

$c_1 = \frac{65 - 48}{20}$ $\quad c_1 = \frac{17}{20}$.

Final solution is

$a_n = \frac{17}{20}(-1)^n + \frac{12}{5}(4)^n - \frac{9}{4} \cdot 3^n$.

3. Solve $a_n - 7a_{n-1} + 10a_{n-2} = 4^n$

Sol. Given that

$a_n - 7a_{n-1} + 10a_{n-2} = 4^n$

L.H.S

$a_n - 7a_{n-1} + 10a_{n-2} = 0$.

Characteristic eqn is

$r^2 - 7r + 10 = 0$

$r^2 - 2r - 5r + 10 = 0$

$r(r-2) - 5(r-2) = 0$

$(r-2)(r-5) = 0$

$r = 2, 5$.

The homogenous solution is

$a_n^{(h)} = c_1(2)^n + c_2(5)^n$.

R.H.S

4^n

For obtaining particular solution we take as $d \cdot 4^n$

Substitute $d \cdot 4^n$ for a_n

$d \cdot 4^n - 7 \cdot d \cdot 4^{n-1} + 10 \cdot d \cdot 4^{n-2} = 4^n$.

$n = 2$

$d \cdot 4^2 - 7 \cdot d \cdot 4^{2-1} + 10 \cdot d \cdot 4^{2-2} = 4^2$

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$$16d - 28d + 10d = 16$$

$$-2d = 16$$

$$d = -8$$

The particular solution is

$$a_n^{(p)} = (-8)\,4^n.$$

$$a_n = c_1(2)^n + c_2(5)^n - 8 \cdot 4^n.$$

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